Scientific and technical factors affecting the setting of Salmonella criteria for raw poultry: a global perspective.

Concerns about foodborne salmonellosis have led many countries to introduce microbiological criteria for certain food products. If such criteria are not well-grounded in science, they could be an unjustified obstacle to trade. Raw poultry products are an important part of the global food market. Import and export ambiguities and regulatory confusion resulting from different Salmonella requirements were the impetus for convening an international group of scientific experts from 16 countries to discuss the scientific and technical issues that affect the setting of a microbiological criterion for Salmonella contamination of raw chicken. A particular concern for the group was the use of criteria implying a zero tolerance for Salmonella and suggesting complete absence of the pathogen. The notion can be interpreted differently by various stakeholders and was considered inappropriate because there is neither an effective means of eliminating Salmonella from raw poultry nor any practical method for verifying its absence. Therefore, it may be more useful at present to set food safety metrics that involve reductions in hazard levels. Such terms as "zero tolerance" or "absence of a microbe" in relation to raw poultry should be avoided unless defined and explained by international agreement. Risk assessment provides a more meaningful approach than a zero tolerance philosophy, and new metrics, such as performance objectives that are linked to human health outcomes, should be utilized throughout the food chain to help define risk and identify ways to reduce adverse effects on public health.